Algorithm 1 Novelty Framework with retrieval, reranking and baseline-normalized scoring

Require: $\mathcal{C}$ (Corpus), $\mathcal{Q}$ (Model generations), $\mathcal{H}$ (Human references), $\mathcal{R}$ (Ranker), $\mathcal{S}$ (Re-ranker), K (Set of chunk sizes $\{k_1, k_2, \dots\}$)
// Stage 1: Identification of Local Candidate Pools
for each document $d \in \mathcal{Q} \cup \mathcal{H}$ **do**
  $P_d \leftarrow \mathcal{R}(d, \mathcal{C}, n)$ ▷ Retrieve n passages once per document using Eq. 1
end for
// Stage 2: Multi-Scale Evaluation and Calibration
for each chunk size $k \in K$ **do**
  $\text{Scores}_H \leftarrow \emptyset, \text{Scores}_Q \leftarrow \emptyset$
// Determine the "Semantic Noise Floor" for this configuration
  for each document $h \in \mathcal{H}$ **do**
    $Chunks_h \leftarrow \text{chunk}(h, k)$
    $S_h \leftarrow \{\max_{p \in P_h} \mathcal{S}(c, p) \mid c \in Chunks_h\}$
    $\text{Scores}_H \leftarrow \text{Scores}_H \cup S_h$
  end for
  $\mu_H^{(k)} \leftarrow \text{mean}(\text{Scores}_H)$ ▷ Stable calibrator for domain and length k
// Stage 3: Calculate Calibrated Similarity Scores for the Model
  for each document $q \in \mathcal{Q}$ **do**
    $Chunks_q \leftarrow \text{chunk}(q, k)$
    $S_q \leftarrow \{\max_{p \in P_q} \mathcal{S}(c, p) \mid c \in Chunks_q\}$
    $R_q \leftarrow \{s / \mu_H^{(k)} \mid s \in S_q\}$ ▷ Normalize by human distribution
    $\text{Scores}_Q \leftarrow \text{Scores}_Q \cup R_q$
  end for
  $N^{(k)} \leftarrow \text{median}(\text{Scores}_Q)$ ▷ Final novelty profile for chunk size k
end for
return $\{(k, N^{(k)}) \mid k \in K\}$ ▷ The novelty profile of the model

The framework is summarized in Algorithm 1. We describe each stage in the following:

1. **Local Candidate Pool Identification.** For every generation in the model distribution $q \in \mathcal{Q}$ and corresponding baseline from the human distribution $h \in \mathcal{H}$, first identify a local candidate pool D within the pretraining corpus $\mathcal{C}$. To this end, a coarse-grained ranker $\mathcal{R}$ retrieves the top-n passages that exhibit the highest potential semantic overlap with the full document. This stage acts as a high-recall filter, ensuring that any potential semantic neighbors are captured once per document, providing a computationally efficient search space for subsequent multi-scale analysis.

2. **Multi-Scale Semantic Re-ranking.** To distinguish between short-range lexical reuse and long-range compositional novelty, evaluate text at multiple chunk sizes k. For a given k, decompose the generation and its corresponding candidate pool into smaller fragments. A high-precision re-ranker $\mathcal{S}$ then computes the maximum similarity between each chunk and its respective pool. This yields a set of *raw similarity scores* for the entire distribution. This two-stage process ensures that the scoring is fine-grained and robust to paraphrasing while remaining tractable at the scale of trillion-token corpora.

3. **Calibrated Distributional Normalization.** As human and model outputs differ in length and structure, a 1:1 element-wise comparison between individual chunks is often impossible: Given one human baseline text for each model generation, they ultimately are split into a different number of chunks for small enough k. Instead, we use the human distribution $\mathcal{H}$ to establish a stable, domain-specific *calibration constant* $\mu_H^{(k)}$ for each chunk size. This constant represents the semantic noise floor, i.e., the level of similarity expected from novel, held-out human text within that domain, configuration, and chunk size. Further motivation

and details on the baseline calibration are provided in Appendix C.

The final *calibrated similarity score* for the model distribution is calculated by normalizing the model's raw scores by the semantic noise floor constant. By aggregating these ratios, we arrive at a *novelty profile* that allows for rigorous comparison across different model architectures or training regimes. This normalization step ensures that any observed trends are not artifacts of the retrieval pipeline or domain-specific redundancies, but true reflections of the model's divergence from natural human patterns of information reuse.

4 Experiments

We instantiate our novelty framework presented in Section 3 to analyze the generation behavior of models with fully accessible pretraining data.

4.1 Experimental Setup

Models and Data ($\mathcal{Q}$ and $\mathcal{C}$). We conduct our analysis on the SmolLM (Allal et al., 2024) and SmolLM2 (Allal et al., 2025) families licensed under the Apache-2.0 license. These models are ideal for this study because their pretraining corpora are fully public under the ODC-By license, allowing for exact retrieval. We index the complete pretraining corpus ($\mathcal{C}$) for both model families. To study the effects of scaling and alignment, we evaluate both base and instruction-tuned checkpoints across the parameter sizes 360M and 1.7B.

Retrieval Instantiation ($\mathcal{R}$ and $\mathcal{S}$). We instantiate the coarse-grained ranker $\mathcal{R}$ using L2-normalized GIST embeddings (Solatorio, 2024) indexed via FAISS (Douze et al., 2024). We chose GIST for its high efficiency-to-performance ratio on the MTEB leaderboard (Muennighoff et al., 2023) at the time of the experiments. For candidate generation (Stage 1 in Algorithm 1), we retrieve $n = 100$ document chunks, which suffices for capturing the most similar corpus document in the vast majority of cases, as validated in Appendix D. We instantiate the re-ranker $\mathcal{S}$ using ColBERTv2 (Santhanam et al., 2022). ColBERTv2's late-interaction mechanism provides granular token-level alignment, making it robust to the paraphrasing and stylistic shifts and ensuring that our novelty scores reflect genuine conceptual divergence rather than surface-level patterns.

Scale of Analysis. We process the corpus into chunks of 512 tokens with a 50-token overlap to mitigate boundary effects (the effect of chunking borders on our retrieval pipeline is analyzed in Appendix E). This results in ∼20 TB of embeddings and indices, which we publish for reproducibility[1]. We perform the analysis across a range of chunk sizes $k \in \{50, 100, \ldots, 500\}$ to measure how generation length affects novelty scores.

4.2 Natural Generation Novelty

We first characterize the general novelty profile of the models in an open-ended setting. Inspired by Merrill et al. (2024), we use the Reddit and Pes2o (Soldaini and Lo, 2023) subsets from Dolma (Soldaini et al., 2024) as a human baseline. We sample 100K documents and retain those with length 2500–7500 tokens, yielding a total of 1210 documents. Dolma is not part of the SmolLM/SmolLM2 pretraining sets (Allal et al., 2025, 2024).

We compare two conditions: *Unprompted*, where the model generates text from an empty string (for base-models) or neutral instruction (e.g. "Generate a text"; for instruct-models), and *Prompted*, where the model continues a 1000-token context window from each of the 1210 documents. Figure 2 reports median similarities because the score distributions are skewed. We further discuss the score distributions in Appendix F.

Not providing context reduces novelty, especially for short outputs. Figure 2 shows that models prompted without context (right plots) achieve higher calibrated similarity scores across chunk sizes than context-conditioned generations (left plots), for both SmolLM and SmolLM2. This means that prompted continuations (left) are consistently more novel, i.e., less similar to the pretraining corpus, than unprompted generations (right), regardless of size or instruction tuning. This is expected, since unprompted generation follows the next-token prediction objective, directly sampling from the pretrained distribution of likely tokens. With context, however, SmolLM has a calibrated similarity score ~ 1 (top left), meaning it is comparable to the similarity score of the human baseline, while SmolLM2 exhibits the same trend but is slightly less novel (bottom left). These findings reflect how conditioning narrows the topical space, whereas unprompted generation more directly mir-

[1] https://huggingface.co/datasets/stai-tuebingen/faiss-smollm

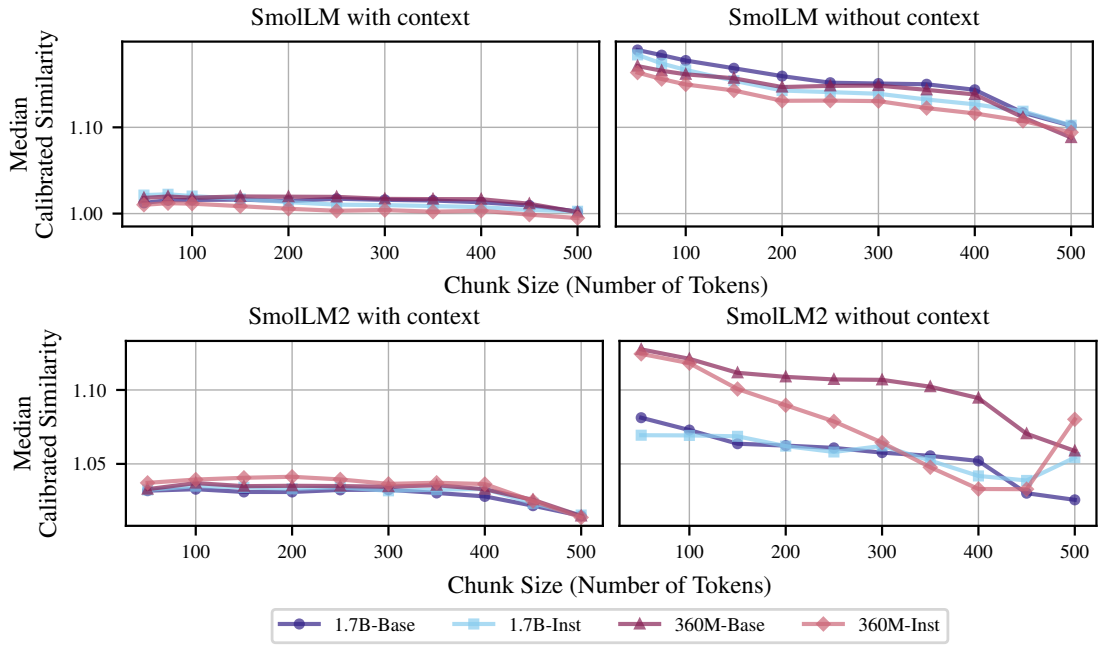

Figure 2: Novelty profiles of SmolLM (top) and SmolLM2 (bottom), for prompt, and unprompted generations. Higher similarity indicates lower novelty.

rors the pretraining data distribution. This forms a contrast to prior observations by (Padmakumar et al., 2025), who did not observe a clear increase in novelty with varying prompting methods.

Novelty increases with sequence length in unprompted generation. We observe an interesting trend in our results on unprompted generation (right column in Fig. 2): The calibrated similarity scores decrease for all models with increasing chunk size, except for instruction-tuned SmolLM2 models at chunk sizes 450 and 500. Without context in the prompt, novelty grows with longer outputs. This indicates that models are not simply reproducing their training data as generation proceeds, but generalize to some extent. Notably, this trend holds across model sizes and architectures.

4.3 Analyzing Domain-Specific Novelty and Instruction Tuning

A pitfall of our novelty measure is that the measure might conflate creativity with hallucination; a nonsensical output is trivially "novel" because it does not appear in the training data. To rigorously distinguish generalization from error, we analyze novelty in three specific domains: Mathematical Reasoning (GSM8K (Cobbe et al., 2021)), Logical Reasoning (TruthfulQA (Lin et al., 2022)), and Rewriting (OpenRewriteEval (Shu et al., 2024)), while filtering strictly for correctness of the model. We include only GSM8K/TruthfulQA samples with perfect accuracy and Rewriting samples with ROUGE-L ≥ 0.25. Dataset sizes are reported in Appendix B. We evaluate TruthfulQA with Gao et al. (2024), GSM8K with Habib et al. (2023), and use a custom script for OpenRewriteEval. Results appear in Figure 3. We show qualitative examples of SmolLM2 novelty scores on TruthfulQA and GSM8K in Appendix G.

Domain constraints dictate novelty. Figure 3 reveals that novelty is highly domain-dependent. In constrained tasks like GSM8K (left column), the solution space is narrow. Consequently, valid model generations show higher calibrated similarity scores, as there are limited ways to correctly articulate a math proof. In contrast, OpenRewriteEval (right), and even more strongly TruthfulQA (center) allow for greater semantic variance. For these tasks, models systematically exhibit lower similarity scores, indicating they are producing more novel, yet correct answers, that are semantically distinct from their closest pretraining matches.

Smaller SmolLM2 models are more novel than larger ones. Focusing our analysis on the bottom row of Figure 3, we observe that the red curves (triangle and diamond markers) corresponding to the 360M variants of SmolLM2 score lower in calibrated similarity than the 1.7B variants (blue